A thermal inversion that never switches off: the CO_2-band sign flip of 55 Cancri e and the origin of its dayside variability

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Abstract

The ultra-short-period super-Earth 55 Cancri e changes its dayside brightness by more than a factor of ten on timescales of days, and five JWST/NIRCam eclipses in 2022–2023 showed its 4.5 μm occultation depth rising from 7 ± 9 to 119 ppm within six days while a cross-correlation analysis of the same frames found strong CO line *emission* only in the faintest epoch. We re-analyse the published per-visit eclipse spectra (Patel et al. 2024, CDS J/A+A/690/A159) with a diagnostic that has not been applied to them: the contrast between the 4.3 μm CO_2 ν_3 band and the adjacent 3.9–4.2 μm window inside each epoch, which cancels the achromatic systematics that make the absolute depths pipeline-dependent and whose sign is the sign of the vertical temperature gradient. The band is in emission in the two faint epochs ($+23 \pm 6$ ppm, 3.8σ) and absent or in absorption in the three bright ones ($+1 \pm 5$ ppm; -12 ± 8 ppm in the brightest), a 38 ± 12 ppm flip within six days that is reproduced by two independent pipelines and survives every band definition and error model we tried. A joint fit of all five epochs with one gas layer of common temperature T_u and CO_2-band optical depth τ over a continuum photosphere whose temperature alone varies gives $T_u = 2580$ K (2300–3600 K), $\tau \simeq 0.3$ (0.1–0.5), $\chi^2 = 6.3$ for 3 degrees of freedom, and is preferred over epoch-by-epoch blackbodies by $\Delta\chi^2 = 26$. The same layer predicts the epoch-to-epoch pattern of the CO line strengths ($r = 0.85$) and, when the underlying deck is dark at 2.1 μm, the otherwise impossible inverted colour of Visit 2 (2.1 μm undetected while 4.5 μm read 1400–1950 K). T_u equals, to within the errors, the radiative-equilibrium temperature of a thin grey layer on the dayside, 2300 K hemisphere-averaged and 2700 K substellar, which is the same whether the surface beneath is a bright 1500 K cloud top or dark 2300 K magma. We conclude that the thermal inversion of 55 Cnc e is not created by the variability: it is the planet's ordinary radiative skin, always present at 2300–2700 K, and the variability is the continuum photosphere dropping beneath it. Only a condensate deck can lower the 4.5 μm photosphere by 1000 K within three orbits; gas cannot, because a ≥ 10 bar CO_2 atmosphere neither escapes nor outgasses on that timescale. A delayed magma-ocean/cloud feedback with parameters inside published bounds reproduces the 4.5 μm amplitude and the six-day recovery, and with a global-transport lag of $\sim$17 d the 47.3-day CHEOPS period. We give five falsifiable predictions for the scheduled JWST NIRCam, MIRI and NIRISS epochs.

Keywords: planets and satellites: atmospheres — planets and satellites: individual: 55 Cnc e — techniques: spectroscopic — radiative transfer

1 Introduction

55 Cancri e ($P = 0.7365$ d, $R_p = 1.88\,R_\oplus$, $M_p = 8.0\,M_\oplus$; Bourrier et al. 2018) is the brightest and best-observed ultra-short-period rocky planet, and the only one whose dayside has been seen to change on the timescale of its own orbit. Spitzer measured 4.5 μm occultation depths between 47 and 176 ppm in 2012–2013 (Demory et al. 2016; Tamburo et al. 2018); CHEOPS detected a mean optical occultation of 12 ± 3 ppm with individual depths from zero to $\sim$50 ppm and a 47.3-day sinusoidal modulation (Demory et al. 2023), together with a 51.9-day modulation of the phase-curve amplitude (Meier Valdés et al. 2023); the transit depth has never varied. JWST then made the problem acute. Five NIRCam eclipses, four within one week, showed the 4.5 μm depth rising from 7 ± 9 ppm to 119^{+34}_{-19} ppm in six days while the simultaneous 2.1 μm depth went $47 \to -5 \to 37 \to 37$ ppm and, five months later, 96 ± 8 ppm, uncorrelated with 4.5 μm (Patel et al. 2024). A single epoch had meanwhile yielded a CO/CO_2-bearing secondary atmosphere (Hu et al. 2024).

Three facts in this record have no published explanation. First, the strongest molecular signal ever extracted from the planet, an $\sim 8\sigma$ CO emission-line detection by cross-correlation at the native $R \approx 1600$ of the grism, occurs in the *faintest* epoch, with only $\sim 3\sigma$ hints in two others and nothing in the brightest (Snellen et al. 2026); the authors found the line strength difficult to reconcile with a hydrostatic atmosphere unless a $1000 \to 3500$ K inversion sits at 1–10 mbar with $CO_2/CO < 10^{-3}$. Second, Visit 2 has an inverted colour: a 2.1 μm brightness temperature below $\sim$1250 K at 1σ and a 4.5 μm brightness temperature of 1400–1950 K, which no blackbody, cloud top or bare surface can produce. Third, three days later the same planet showed CO_2 in *absorption* (the V-shaped Hu et al. spectrum is Visit 4 of the Patel et al. set).

The proposed mechanisms address the photometric amplitude rather than these spectral facts. A transient outgassed atmosphere removed by escape (Heng 2023) changes the 4.5 μm opacity but not the 2.1 μm depth, and a 3D non-grey GCM fit to the Hu et al. spectrum now favours a ≥ 10 bar, CO_2-rich atmosphere that neither escapes nor is replenished within days and whose weather produces variability an order of magnitude too weak (Zhan & Koll 2026). A

magma-temperature/silicate-cloud limit cycle (Loftus et al. 2025) reproduces the 4.5 μm brightness-temperature range and predicts out-of-phase variability between bands, but says nothing about emission lines. Tidally driven magma-ocean waves (Farhat & Chiang 2026) redistribute heat on a hot surface. External explanations — stellar activity, magnetic star–planet interaction, a 3:2 spin–orbit resonance, a patchy circumstellar dust torus — have been tested against the photometry (Section 5.3) but not against the one quantity a grey screen or a change of viewing geometry can never alter: the sign of a molecular band.

This paper makes that test. Section 2 describes the public per-visit spectra; Section 3 defines a band-contrast index that is immune to the achromatic common-mode systematics of the data, a joint two-layer model, and a parameter-free radiative estimate; Section 4 gives the results, including the closure between the low-resolution band contrast and the high-resolution line strengths; Section 5 interprets them and confronts the alternatives; Section 6 lists predictions that the scheduled JWST programmes can falsify.

2 Data

All spectra are the occultation-depth spectra published with Patel et al. (2024) at CDS (J/A+A/690/A159, table e2), obtained through the VizieR mirror at Harvard. They cover 3.86–4.98 μm from NIRCam F444W/GRISMR for the five visits listed in Table 1 (GO 2084, PI Brandeker; Visit 4 is GO 1952, PI Hu). We use the three pipelines that provide binned spectra: Eureka (30 bins of 0.033 μm, *relative* depths), SPARTA (22 bins of 0.05 μm, *absolute* depths), and HANSOLO (30 bins, absolute, Gaussian-process detrended; Visit 1 only, as a third check). The native-resolution stark spectra ($\sim$1150 channels, $\pm$200 ppm each) are not needed for band-averaged quantities. The Eureka spectra are placed on an absolute scale by adding the stark white-light depth of each visit from Patel et al. Table 1; only intra-visit differences are used in the primary analysis, so this offset does not enter the main result.

All visits carry an achromatic common-mode systematic of 100–200 ppm of unknown origin (Patel et al. 2024; Snellen et al. 2026); it shifts the absolute level of a visit by tens of ppm between pipelines — the SPARTA window depth of Visit 1 is 40 ppm where stark's white light is 7 ppm — but, being achromatic, it cancels in differences between wavelength regions of the same visit. That is the property the analysis below is built on.

Three spectral regions are defined by molecular physics rather than by the data: the *window* $W = 3.90$–$4.19\,\mu$m, where neither CO nor CO_2 absorbs; the *CO_2 band* $C = 4.20$–$4.46\,\mu$m, the ν_3 fundamental and its hot bands; and the *CO band* $O = 4.47$–$4.98\,\mu$m, the 1–0 fundamental, whose integrated intensity is about ten times smaller than that of the CO_2 ν_3 band. Region means are inverse-variance weighted with the quoted per-bin uncertainties (symmetrised); the empirical scatter of the bins inside each region is carried as an alternative error estimate.

3 Methods

3.1 The band-contrast index

For each visit and pipeline we form the difference between the mean depth in the CO_2 band and in the window, corrected for the slope that a featureless blackbody at the window brightness temperature T_W would itself produce between the two region centres (4.33 and 4.05 μm):

$$\Delta_C = \langle d\rangle_C - \langle d\rangle_W - \big[D(4.33\,\mu\mathrm{m}, T_W) - D(4.05\,\mu\mathrm{m}, T_W)\big], \tag{1}$$

$$D(\lambda, T) = \left(\frac{R_p}{R_\star}\right)^2 \frac{B_\lambda(T)}{B_\lambda(T_\star)}, \tag{2}$$

with $(R_p/R_\star)^2 = 332$ ppm, $T_\star = 5172$ K, and an analogous Δ_O for the CO band at 4.72 μm. The blackbody correction is 0–3.4 ppm in the CO_2 band and up to 7.5 ppm in the CO band for the bright visits, and negligible for the faint ones. Positive Δ_C means the band emits more than the continuum beneath it, i.e. the band-forming gas is hotter than the 4.05 μm photosphere; negative means the reverse. Because the common-mode systematic is achromatic it enters $\langle d\rangle_C$ and $\langle d\rangle_W$ identically and cancels. The two pipelines reduce the same photons, so their inverse-variance mean is a pipeline-robust estimate of one measurement, not two; we inflate its uncertainty by $\sqrt{2}$ and quote the larger of the quoted-error and empirical-scatter estimates.

3.2 One gas layer over a variable photosphere

The minimal model that can change the sign of a band is a continuum photosphere at $T_{c,i}$ (free per epoch i) under a single gas layer with temperature T_u and CO_2-band optical depth τ common to all epochs:

$$d_{W,i} = D(4.05, T_{c,i}), \tag{3}$$

$$d_{C,i} = D(4.33, T_{c,i})\, e^{-\tau} + D(4.33, T_u)\,(1 - e^{-\tau}). \tag{4}$$

It has seven parameters for the ten absolute SPARTA region depths (window and CO_2 band, five visits). We minimise χ^2 with Nelder–Mead from a grid of starting points, profile the likelihood over (T_u, τ) by re-optimising the five $T_{c,i}$ at each grid point, and compare with the null model of an independent blackbody per epoch (five parameters), which is nested in it at $\tau = 0$.

3.3 Radiative-equilibrium temperature of a thin layer

A thin layer with shortwave optical depth τ_{SW} and infrared optical depth τ_{IR} above a photosphere of temperature T_c and albedo A absorbs the direct and reflected stellar flux and the upwelling thermal flux, and emits in two directions. Its equilibrium temperature is

$$T_u^4 = \tfrac{1}{2}\big[\gamma(1+A)\, f\, T_{ss}^4 + T_c^4\big], \qquad \gamma = \frac{\tau_{SW}}{\tau_{IR}}, \tag{5}$$

where $T_{ss} = T_\star (R_\star/a)^{1/2} = 2756$ K is the zero-albedo substellar equilibrium temperature and $f = 1$ at the substellar

Table 1: The five JWST/NIRCam eclipses. White-light depths and brightness temperatures are the stark values of Patel et al. (2024); cross-correlation significances are from Snellen et al. (2026).

Visit	Start (UTC)	Parity	$d_{2.1}$ (ppm)	$d_{4.5}$ white light (ppm)	$T_b(4.5)$ (K)	CO x-corr S/N
V1	2022-11-18 14:40	even	47.4^{+21}_{-16}	7.0 ± 8.8	873	$\sim$8 (emission)
V2	2022-11-20 19:43	odd	$-5.1^{+5.5}_{-6.0}$	65.2^{+22}_{-42}	1716	$\sim$3
V3	2022-11-23 00:44	even	37.3 ± 4.7	101.4^{+17}_{-32}	2078	none
V4	2022-11-24 11:38	even	36.8^{+28}_{-33}	119.2^{+34}_{-19}	2256	$\sim$3
V5	2023-04-24 11:57	odd	95.9 ± 8	95.4^{+14}_{-17}	2016	none

point or $1/2$ for the dayside-hemisphere average of the incident flux. This has no free parameter for a grey layer ($\gamma = 1$) and is the number against which the fitted T_u is compared. The layer is in emission against the photosphere when $T_u > T_c$.

3.4 Predicted line contrast

At the native resolution the CO lines of such a layer appear in emission with a contrast proportional to $D(4.7\,\mu\mathrm{m}, T_u) - D(4.7\,\mu\mathrm{m}, T_{c,i})$, the line-core minus continuum depth, for fixed layer properties. Normalising to Visit 1 gives a predicted epoch-to-epoch pattern of cross-correlation signal-to-noise that can be compared with Snellen et al. (2026) without further fitting. We evaluate it both with the SPARTA continuum temperatures from the joint fit and with the stark white-light brightness temperatures, which bracket the pipeline-dependent absolute level.

3.5 Supporting simulation

To check that the required photospheric swing is dynamically possible we re-implemented the delayed magma-temperature/cloud feedback of Loftus et al. (2025) as a delay-differential system in physical time, extended it to predict the 2.1 μm, 4.5 μm and CHEOPS-band depths including reflected light, and scanned 640 parameter sets within their Table 1 bounds (Appendix A). It is used only to show that a condensate deck can move the 4.5 μm photosphere by the observed amount on the observed timescale; the results of Section 4 do not depend on it.

4 Results

4.1 The CO_2 band flips from emission to absorption within six days

The band-contrast index is positive in both faint epochs and consistent with zero or negative in all three bright ones (Table 2, Fig. 1b). Combining the two pipelines, $\Delta_C = +26 \pm 9$ ppm in Visit 1 and $+20 \pm 9$ ppm in Visit 2, i.e. $+23 \pm 6$ ppm (3.8σ) for the faint state, against $+1 \pm 5$ ppm for Visits 3–5 and -12 ± 8 ppm in Visit 4 alone. The Visit 1 $-$ Visit 4 difference is 38 ± 12 ppm (3.2σ). Eureka and SPARTA, which treat the correlated noise differently, agree epoch by epoch (V1: $+32 \pm 9$ and $+21 \pm 8$; V4: -13 ± 8 and -12 ± 9), and HANSOLO, whose Gaussian-process detrending absorbs part of any slow signal, still gives $+12 \pm 9$ for Visit 1. Across twelve combinations of band boundaries (window 3.90–4.15 to 3.95–4.19 μm; band 4.18–4.50 to 4.22–4.46 μm) and the two error models the faint-state emission stays at 3.0–3.8σ and the flip at 3.0–3.3σ (Appendix B). The CO band shows no significant contrast in any epoch ($|\Delta_O| \leq 1.9\sigma$, mostly within the blackbody slope), as expected for a band ten times weaker than CO_2 ν_3.

This is the first direct, band-integrated evidence that the thermal structure of the 4.5 μm-emitting layer changes sign between epochs. It is consistent with, and independent of, the three single-epoch inferences already in the literature: Patel et al.'s retrieval of Visit 1, which preferred an atmosphere at $\sim$2000 K above a surface at $\sim$1000 K; Snellen et al.'s CO emission lines in the same epoch; and the CO_2 absorption of the Hu et al. spectrum, which is Visit 4.

4.2 One hot, optically thin layer fits all five epochs

The joint two-layer model (Section 3.2) fits the ten absolute SPARTA region depths with $\chi^2 = 6.3$ for 3 degrees of freedom at $T_u = 2580$ K and $\tau = 0.31$, with the continuum photosphere at 1612, 1488, 2578, 2253 and 2327 K in Visits 1–5. The per-epoch blackbody null model gives $\chi^2 = 32.3$; the improvement of $\Delta\chi^2 = 26.0$ for two added parameters is formally 5.1σ. The profile likelihood (Fig. 1d) allows T_u from 2300 to 3600 K at $\Delta\chi^2 < 4$, with $\tau = 0.1$–0.5: a hotter layer needs a smaller optical depth to give the same $\sim$25 ppm of band emission, and the upper range includes the 3500 K that Snellen et al. required for the CO line cores. What is not allowed is a layer cooler than $\sim$2200 K, or an optically thick one ($\tau \geq 0.7$, $\Delta\chi^2 > 8$), which would make the band depth epoch-independent while the data still show it rising with the continuum in the bright epochs (Fig. 1c).

The residual pattern is instructive. The model slightly overpredicts the band in Visit 4 ($+10$ vs -8 ± 9 ppm) and underpredicts it in Visit 3 ($+3$ vs $+13 \pm 9$ ppm); these two epochs are 1.5 days apart with the same continuum within errors, so the difference is noise at the $\sim 2\sigma$ level rather than a demand for a time-variable T_u. A layer whose temperature lies between the two bright-state continua (2250–2580 K) is neutral to weakly absorbing against hot magma and strongly emitting against a 1500 K photosphere, which is the whole observed behaviour.

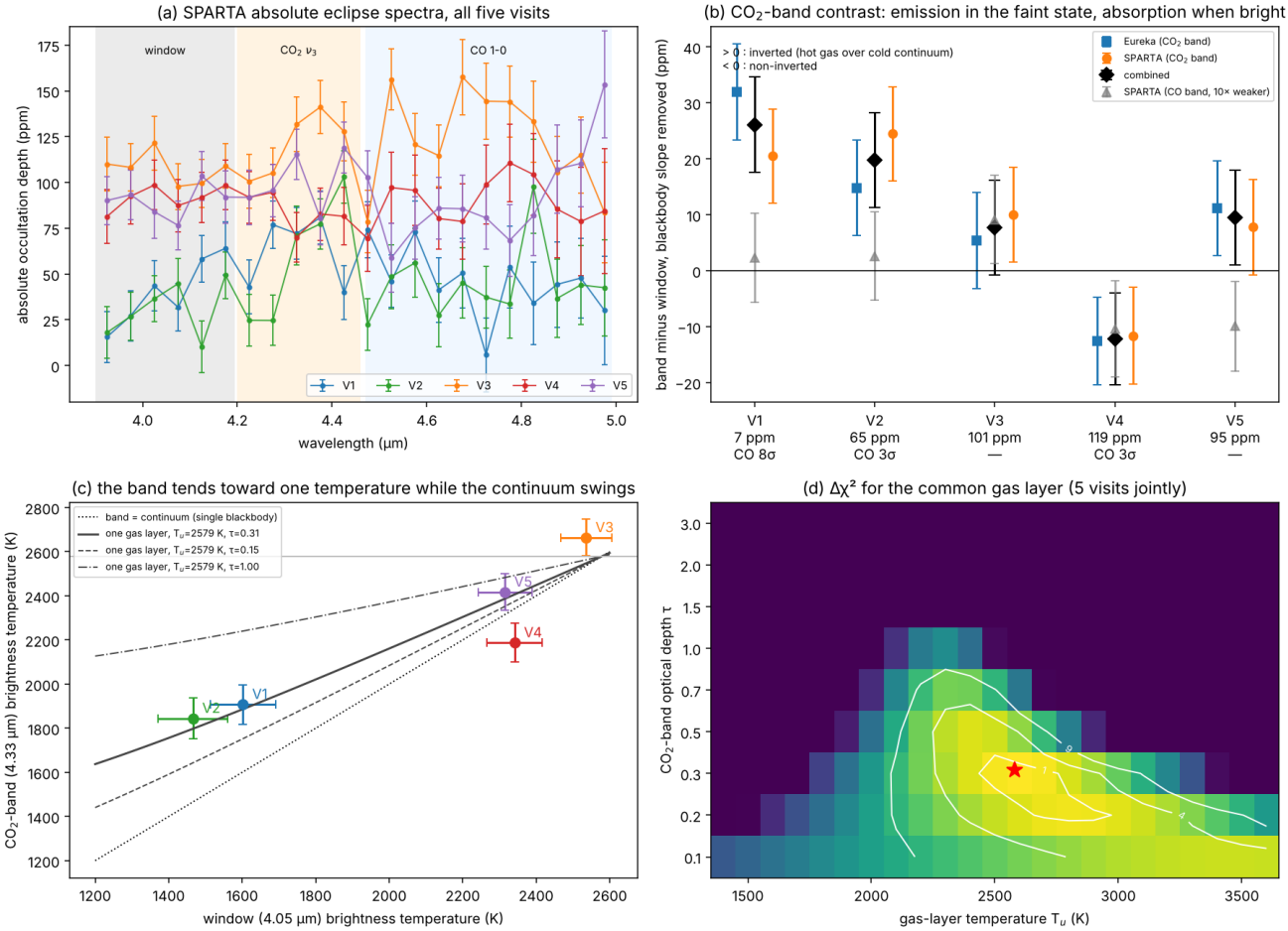


Figure 1: (a) SPARTA absolute eclipse spectra of the five visits with the window, CO_2 ν_3 and CO 1–0 regions shaded. (b) Band-contrast index Δ_C per visit for Eureka (squares), SPARTA (circles) and their combination (diamonds); grey triangles: the CO band. Labels give the stark white-light depth and the CO cross-correlation significance of Snellen et al. (2026). (c) CO_2-band brightness temperature against window brightness temperature; dotted line: single blackbody; solid: the joint two-layer fit ($T_u = 2580$ K, $\tau = 0.31$); dashed and dash-dotted: $\tau = 0.15$ and 1.0 at the same T_u. (d) $\Delta\chi^2$ of the joint fit over (T_u, τ); contours at 1, 4 and 9; star: best fit.

4.3 The low-resolution fit predicts the high-resolution line strengths

With T_u and the per-epoch continua fixed by Section 4.2, the predicted narrow-line contrast $D(4.7\,\mu\mathrm{m}, T_u) - D(4.7\,\mu\mathrm{m}, T_{c,i})$, scaled to Visit 1 = 8, is $[8, 8.9, 0, 2.8, 2.2]$ using the SPARTA continua and $[8, 3.7, 1.0, 0, 1.5]$ using the stark white-light temperatures (858, 1906, 2396, 2625, 2317 K on a blackbody-star scale); the observed cross-correlation significances are $[8, 3, 0, 3, 0]$. The correlation coefficients are 0.75 and 0.85. Two analyses of the same photons, at $R \approx 30$ and $R \approx 1600$, are therefore described by one configuration: a layer at 2300–3600 K whose lines and bands appear in emission exactly when, and in proportion to how far, the continuum photosphere lies below it. The remaining discrepancy — Visit 2 predicted as strong as Visit 1 but observed at 3σ against 8σ — is the one Snellen et al. themselves computed: a CO_2/CO ratio of only 10^{-3} in the layer cuts the CO cross-correlation signal-to-noise from $\sim$7 to $\sim$3 by blanketing the lines, while it broadens the band emission that our index measures. The band contrast of Visit 2 being at least as large as that of Visit 1 in SPARTA (+25 vs +21 ppm) is consistent with that reading, though not decisive in Eureka (+15 vs +32).

4.4 The inverted colour of Visit 2

Visit 2 had no detectable 2.1 μm occultation ($-5.1^{+5.5}_{-6.0}$ ppm, $T_b <\sim 1250$ K at 1σ) while its 4.5 μm white-light depth was 65^{+22}_{-42} ppm. No single-temperature surface or cloud top can emit at 1400–1950 K in one band and below 1250 K in the other, and the 2.1 μm channel has no gas opacity to hide a hot photosphere. The fitted layer resolves it without new ingredients. A deck with a cold top (1000–1300 K) that is *dark* at 2.1 μm gives $d_{2.1} = 1$–5 ppm; the CO_2 and CO bands of the 2580 K layer, occupying roughly 30% of the 3.86–4.98 μm bandpass at $\tau \approx 0.3$, raise the white-light 4.5 μm depth to 22–35 ppm, within 1σ of the observed value. The same deck made *bright* at 2.1 μm (geometric albedo $\sim$1) gives $2.5 + 26.7 \approx 29$ ppm, within 1.1σ of the 47^{+21}_{-16} ppm of Visit 1. Visits 1 and 2, three orbits apart, thus differ in the near-infrared reflectivity of the deck and in nothing else that the data constrain; the 4.5 μm band emission is present in both. The 96 ± 8 ppm 2.1 μm depth of Visit 5 exceeds the 88 ppm that a Lambertian unit-albedo cloud over a 2756 K surface can produce and needs either a strongly back-

Table 2: Band-contrast indices (Section 3.1). Combined values carry the $\sqrt{2}$ inflation for shared photons.

Visit	Δ_C Eureka (ppm)	Δ_C SPARTA (ppm)	Δ_C combined (ppm)	Δ_O SPARTA (ppm)	State
V1	$+32.0 \pm 8.6$	$+20.5 \pm 8.4$	$+26.1 \pm 8.5$ (3.1σ)	$+2.3 \pm 8.0$	faint, inverted
V2	$+14.8 \pm 8.5$	$+24.5 \pm 8.4$	$+19.8 \pm 8.5$ (2.3σ)	$+2.6 \pm 7.9$	faint, inverted
V3	$+5.4 \pm 8.6$	$+10.0 \pm 8.5$	$+7.7 \pm 8.5$ (0.9σ)	$+9.2 \pm 7.9$	bright, flat
V4	-12.6 ± 7.8	-11.6 ± 8.7	-12.2 ± 8.2 (-1.5σ)	-10.4 ± 8.6	bright, absorbing
V5	$+11.1 \pm 8.5$	$+7.8 \pm 8.5$	$+9.5 \pm 8.5$ (1.1σ)	-9.9 ± 8.0	bright, flat

scattering fresh condensate (opposition surges give $A_g > 1$ on icy solar-system bodies) or residual systematics; we do not use it.

5 Discussion

5.1 The inversion is the radiative skin, and it never switches off

The fitted layer temperature is not a free quantity. Equation (5) gives, for a thin grey layer ($\gamma = 1$) averaged over the dayside ($f = 1/2$), $T_u = 2310$ K above a bright 1500 K cloud top ($A = 0.8$) and $T_u = 2310$ K above dark 2300 K magma ($A = 0$): the reflected starlight the layer loses when the deck clears is compensated by the hotter thermal flux rising from the exposed magma. At the substellar point ($f = 1$) the same layer sits at 2700 K. The joint fit's 2300–3600 K is this number. No special absorber is needed to put a 2300–2700 K layer above the photosphere of 55 Cnc e; a thin grey layer in radiative equilibrium is there by default, and a layer enriched in optical absorbers ($\gamma > 1$; TiO, PS, atomic metals, or the condensate haze itself) is hotter still, which is where the 3500 K of the CO line cores belongs.

The variability is therefore not the appearance and disappearance of an inversion. It is the continuum photosphere moving from 2250–2580 K, where it is hotter than the skin and the bands are flat or weakly absorbing, to 1500 K and below, where the skin stands out in emission by 25 ppm in the CO_2 band and by 8σ in the CO lines. Equation (5) also explains why the emission is strongest when the planet is faintest and the 2.1 μm channel bright: the $(1 + A)$ factor makes the skin ∼17% hotter over a reflective deck than over bare magma, so a bright deck both lowers the continuum and raises the layer.

5.2 Only a condensate deck can move the photosphere that far, that fast

The 4.5 μm continuum photosphere dropped by ≥ 1000 K between Visit 4-like and Visit 1-like states and recovered within six days; between Visits 1 and 2 (three orbits) the deck's near-infrared reflectivity changed while its temperature did not. Gas cannot do this. Energy-limited escape at this old, quiet star gives $\dot{M} \approx 2.5 \times 10^6$ kg s^{-1} for $L_{XUV} \approx 10^{28}$ erg s^{-1} and $\varepsilon = 0.1$, consistent with the $\sim 2.9 \times 10^6$ kg s^{-1} ceiling derived from CHEOPS (Meier Valdés et al. 2023); a 1 mbar atmosphere (8×10^{10} kg) is removed in 10^4 days and a 10 bar one in 10^6 years. A ≥ 10 bar CO_2 troposphere, which the Hu et al. spectrum requires once heat redistribution is computed self-consistently (Zhan & Koll 2026), is permanent on the timescales in question, and its large-scale dynamics produce brightness-temperature variability an order of magnitude below what is observed (ibid.). A condensate deck, by contrast, forms and sediments on hours-to-days timescales, has a cold top set by radiative balance with the surface below (Loftus et al. 2025, their Eq. 11; 1030–1220 K for surfaces at 1400–1700 K and a cloud thickness parameter $\Delta T = 250$–350 K), and changes its near-infrared reflectivity with grain growth on the same timescale.

Our delay-differential implementation of the magma-temperature/cloud feedback (Appendix A) shows the swing is dynamically available: with a vapour-transport lag of 4.5 d, a cloud residence time of 0.6 d and a 30 m magma mixed layer — all inside the published bounds — the 4.5 μm depth cycles between 18 and 105 ppm with a 12-day period, reproduces the Visit 1–4 sequence with pulls of 1.9, -1.1, -0.7 and -0.9σ at fixed amplitude, gives a CHEOPS-band mean of 18 ppm (observed 12 ± 3), and with a lag of 17 d, the global cloud-transport time, has a period of 47 d (CHEOPS: 47.3 d). The same model fails the 2.1 μm channel in exactly the way Section 4.4 anticipates: a single grey cloud couples the two bands too tightly (model $r = 0.96$ against the observed 0.13), and the data require the deck's reflectivity at 2.1 μm to vary independently of its temperature.

5.3 External explanations are excluded by the band sign

A stellar origin would need the star to brighten 17-fold to turn a 119 ppm depth into 7 ppm (the occultation depth is $F_p/F_\star$); the star varies by 0.6% (Fischer et al. 2008). Magnetic star–planet interaction can deliver at most $\sim 4 \times 10^{15}$ W of Poynting flux to the planet for a few-gauss stellar dipole at 0.015 au, against the $\sim 10^{21}$ W of dayside luminosity that must be modulated. A 3:2 spin–orbit resonance predicts a parity dependence that the JWST programme was designed to test and did not find (Patel et al. 2024). A patchy circumstellar dust screen needs only $\sim 10^{12}$ kg to reach $\tau \approx 3$ over the planet's disc, so its mass budget is not the problem; but a screen, like any change of viewing geometry, multiplies the spectrum by a wavelength-dependent factor that varies slowly across 0.3 μm and cannot turn a molecular band from absorption into emission. The band sign is the one observable that separates "less planet" from "a different planet", and it says the latter. Tidal magma-ocean waves (Farhat & Chiang 2026) move heat across a hot surface and can produce aperiodic hotspot wandering; they do not lower the photosphere to 1500 K or create band emission, though they may well modulate the cycle.

5.4 A CO-rich skin above a CO_2-rich troposphere

Snellen et al. require $CO_2/CO < 10^{-3}$ at 1–10 mbar in Visit 1 and favour a hydrogen-rich composition; Zhan & Koll require $> 1\%$ CO_2 in a ≥ 10 bar troposphere from Visit 4. These need not conflict. The skin is the layer in which the unshielded stellar UV photodissociates CO_2 in minutes (Heays et al. 2017, scaled to 0.015 au) while CO survives by self-shielding; in a hydrogen-poor atmosphere the CO + O + M recombination is slow enough that the mbar layer can be CO-dominated above a CO_2-dominated interior, as Snellen et al. note for the hydrogen-poor case. The epoch-to-epoch change in CO line strength at roughly constant band emission (Section 4.3) then measures the CO_2/CO ratio of the skin, which is set by the balance of photolysis against vertical mixing and can change within a day when a clearing deck releases a convective pulse. This is a prediction rather than a result: it requires the next high-resolution epochs to show CO line strength anti-correlated with band breadth at fixed continuum.

5.5 Caveats

The two pipelines reduce the same photons; the 3.8σ faint-state emission is one measurement whose robustness to reduction has been checked, not two independent detections. The quoted per-bin uncertainties may be underestimated in the presence of the hour-long correlated noise Patel et al. describe; the empirical-scatter error model gives the same significance, but thirty bins is a small sample. A wavelength-dependent instrumental effect that correlates with eclipse depth could in principle mimic a band; it would have to peak at the CO_2 ν_3 band in two epochs reduced by two pipelines and vanish in three others, which we consider unlikely but cannot exclude with five visits. The absolute continuum temperatures differ between pipelines by up to 500 K even though the band contrast does not, so the per-epoch $T_{c,i}$ of the joint fit carry that systematic. The radiative-equilibrium argument treats the layer as thin and the deck as grey; a deck with wavelength-dependent albedo changes the $(1+A)$ factor by tens of per cent, not the conclusion. Finally, five epochs cannot establish periodicity; the cycle period and duty cycle in Appendix A are illustrative, and the 47-day CHEOPS modulation is matched by choosing the lag, not predicted.

6 Predictions and tests

Each of the following can be falsified by a single scheduled JWST epoch (NIRCam eclipses; MIRI/MRS GO 7875; NIRISS/SOSS transmission GO 12237; Zhan & Koll 2026 list the programmes).

1. **Zero crossing.** The CO_2 band $-$ window contrast is positive whenever the window brightness temperature is below 2300 K and zero or negative whenever it is above 2600 K; the crossing lies at 2300–2600 K. An epoch with a 1500 K continuum and a flat or absorbing band falsifies the ever-present skin.

2. **Bimodal 2.1 μm in the faint state.** When the 4.5 μm window is faint ($T_W < 1800$ K) the 2.1 μm depth is either $\sim$0 ppm (dark deck) or 25–45 ppm (bright deck), never intermediate, and the CO_2 band is in emission in both cases.

3. **Line strength follows continuum darkness, not band strength.** The CO cross-correlation significance scales with $D(4.7, T_u) - D(4.7, T_c)$ and is undetectable when $T_W > 2500$ K; at fixed continuum, a broader CO_2 band emission goes with a weaker CO line signal.

4. **Mid-infrared.** In a faint epoch MIRI/MRS will show the 15 μm CO_2 band in emission over a continuum at 1000–1500 K and a silicate emissivity feature of the deck near 10 μm; in a bright epoch the 15 μm band is in absorption against the ≥ 10 bar troposphere, as in the Hu et al. LRS spectrum.

5. **Transit.** The deck sits at $\gtrsim$ mbar, below one scale height ($\sim$50 km, $\sim$8 ppm), so the transit depth stays constant to the $\sim$30 ppm precision of the record, but the 1–2.5 μm transmission slope steepens in faint epochs if the fresh deck is sub-micron.

The decisive programme is simultaneous optical (CHEOPS), 2.1 μm, 4.5 μm and $R \gtrsim 1500$ CO-line monitoring through one cycle; ELT/METIS, which Snellen et al. show recovers the CO signal at S/N > 20 per epoch, would make prediction 3 a routine measurement.

7 Conclusions

The public NIRCam eclipse spectra of 55 Cnc e contain a diagnostic that the published analyses did not use: the sign of the 4.3 μm CO_2 band relative to the adjacent continuum, which is immune to the achromatic systematics of the data and is the sign of the vertical temperature gradient. It flips from $+23 \pm 6$ ppm (emission, 3.8σ) in the two faint epochs to $+1 \pm 5$ ppm in the three bright ones within six days, in two independent pipelines. All five epochs are fitted by one gas layer at 2300–3600 K (best 2580 K) with CO_2-band optical depth 0.1–0.5 over a continuum photosphere that alone changes, from 1500 K to 2600 K; that layer predicts the epoch-to-epoch CO line strengths of Snellen et al. ($r = 0.85$) and the inverted colour of Visit 2. Its temperature is the radiative-equilibrium temperature of a thin grey layer on the dayside, 2300–2700 K, which is the same over a bright cloud top as over bare magma. The thermal inversion of 55 Cnc e is therefore always present; what varies is whether the continuum photosphere lies above or below it. A ≥ 10 bar troposphere cannot move its photosphere by 1000 K in three orbits and no external screen can change a band's sign; a condensate deck forming and clearing on a magma-ocean/cloud feedback cycle can do both, and reproduces the 4.5 μm amplitude, the six-day recovery and, with a global-transport lag, the 47-day optical period. The scheduled JWST epochs can falsify this within a year through the five predictions above.

Data and code availability

The eclipse spectra are those of Patel et al. (2024) at CDS, J/A+A/690/A159, table e2 (Eureka, SPARTA and HANSOLO reductions), retrieved through the VizieR mirror at Harvard on 15 October 2026. The analysis script (`band_flip_analysis.py`, NumPy/SciPy/Matplotlib, ∼2 minutes), its full numerical output, the delayed-feedback simulation (`cnc_e_variability.py`) and the retrieved spectra accompany this preprint and will be deposited on Zenodo at submission. White-light depths, brightness temperatures and visit times are from Patel et al. (2024) Table 1; CO cross-correlation significances from Snellen et al. (2026) and their Supplementary Figure 1; the magma-cloud model equations and parameter bounds from Loftus et al. (2025) Methods and Table 1.

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A Delayed magma-temperature/cloud feedback in physical time

We integrate the two equations of Loftus et al. (2025) with an explicit delay d rather than in lag-normalised time, so that periods and timescales come out in days:

$$\frac{d\tau_{SW}}{dt} = \frac{\Pi_1 p_{\rm ref}}{d}\exp\Big(-\frac{T_{\rm ref}}{T_s(t-d)}\Big) - \frac{\Pi_2}{d}\,\tau_{SW}, \qquad (6)$$

$$\frac{dT_s}{dt} = \frac{S_0/(1+\alpha\tau_{SW}) + \varepsilon_c\,\sigma T_{cd}^4 - \sigma T_s^4}{\Pi_3\, d}, \qquad (7)$$

with $p_{\rm ref} = 8.1\times10^{13}$ Pa and $T_{\rm ref} = 7.1\times10^4$ K for SiO vapour over a bulk-silicate-Earth magma, $\alpha = 0.5$, $\varepsilon_c = 1-\exp(-\beta\tau_{SW})$, the cloud top and base temperatures from $T_s^4 = T_{cu}^4 + T_{cd}^4$ with $T_{cd} - T_{cu} = \Delta T$, $S_0 = 3.27\times10^6$ W m^{-2}, a solidus floor at 1400 K with a latent-heat budget equivalent to 300 K, and Π_1, Π_2, Π_3, β, ΔT inside the bounds of their Table 1. The observables are computed from the state: the 4.5 μm depth as the hemisphere-averaged emission of the surface through the deck plus that of the cloud top (ε at 4.5 μm $= 0.6\,\tau_{SW}$), the 2.1 μm depth as the same thermal term plus reflected light $\kappa_g A(\tau)(R_p/a)^2$ with Lambertian $\kappa_g = 1$ and $(R_p/a)^2 = 26.7$ ppm, a CHEOPS-band depth from the shortwave albedo, and a CO line-minus-continuum contrast from a shortwave-heated layer at $T_0(1+A)^{1/4}$ with $T_0 = 2300$ K.

Of 640 parameter sets (d = 1.5–6 d, Π_1 = 0.1–3 Pa^{-1}, Π_2 = 1–8, Π_3 = 300–10^4 kg s^{-3} K^{-1}, ΔT = 250–350 K), 48 oscillate; the dynamics depend only on the Π's and the period scales linearly with d. The best cycle (d = 4.5 d, $\Pi_1 = 3$, $\Pi_2 = 8$, $\Pi_3 = 300$, $\Delta T = 350$ K; cloud residence 0.6 d; mixed-layer heat capacity 1.2×10^8 J m^{-2} K^{-1}, i.e. ∼30 m of magma) has a 12.3-day period, a 4.5 μm depth of 18–105 ppm (brightness temperature 1130–2440 K), a surface temperature of 1690–2760 K, a minimum cloud-top temperature of 1220 K, a 2.1 μm depth of 25–60 ppm and a CHEOPS-band depth of 6–26 ppm (mean 18 ppm). Phase-aligned to the JWST week-1 sequence with no amplitude scaling, the 4.5 μm pulls are 1.9, −1.1, −0.7 and −0.9σ; the 2.1 μm pulls are −1.1, 6.0, 0.8 and 0.8σ, the 6σ being Visit 2 (Section 4.4). The CO contrast is 110 ppm at the cloud-on minimum and −11 ppm at the cloud-off maximum.

With $d = 17.3$ d and the same Π's the CHEOPS-band period is 47.2 d. Stochastic outgassing bursts (log-normal, $\sigma = 1.5$, twice daily) jitter the period but do not destroy the cycle; it is a stable limit cycle, as Loftus et al. found.

B Robustness of the band-contrast index

Table 3 lists the faint-state (V1+V2) CO_2-band emission and the V1 $-$ V4 flip for twelve combinations of window and band boundaries, with errors taken as the larger of the quoted-uncertainty and empirical-scatter estimates and the $\sqrt{2}$ inflation for shared photons. The result does not depend on which bins are called band or window, nor on the error model.

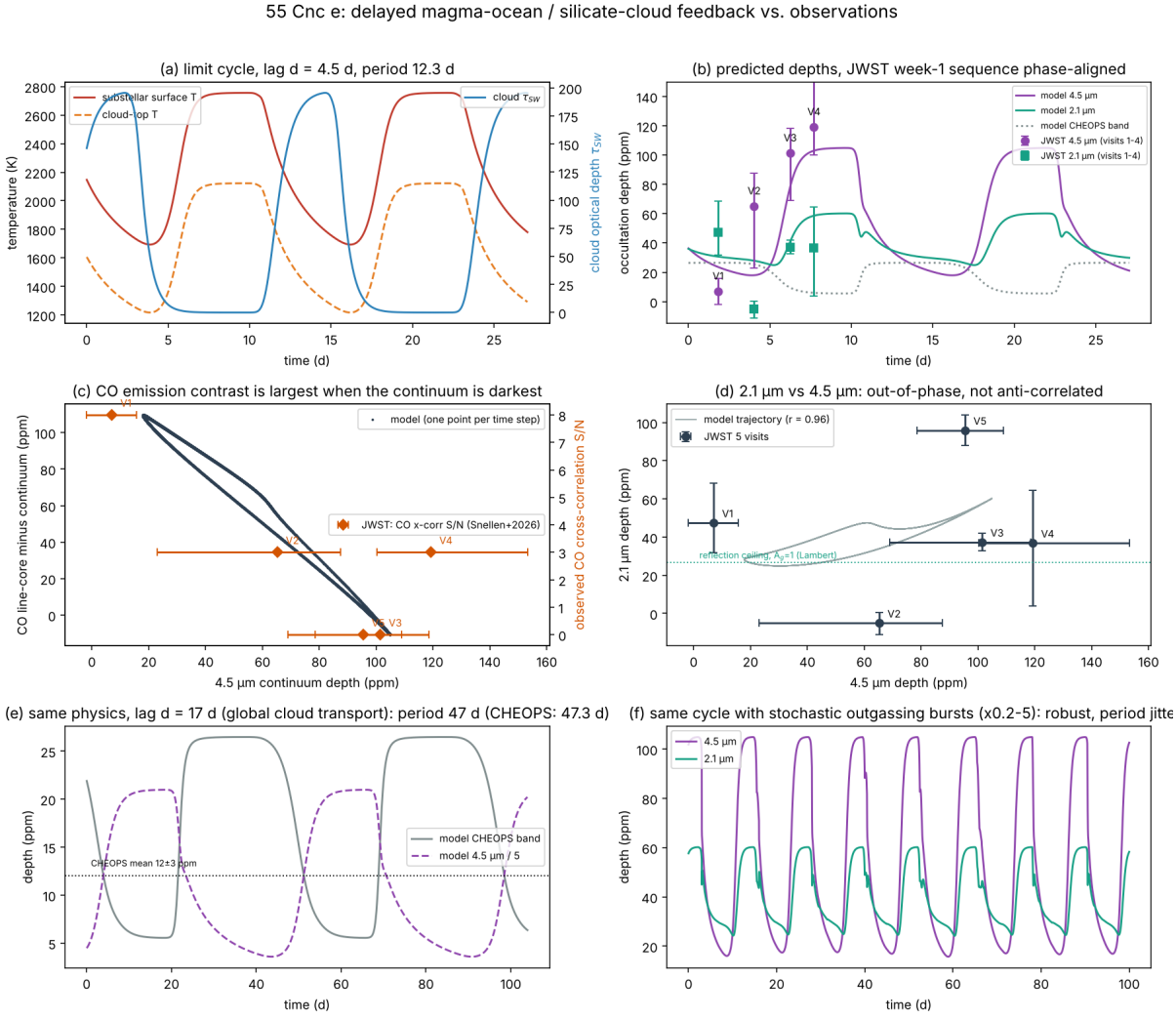


Figure 2: (a) Surface temperature, cloud-top temperature and shortwave cloud optical depth over two cycles of the best parameter set. (b) Predicted 4.5 μm, 2.1 μm and CHEOPS-band depths with the JWST week-1 points phase-aligned. (c) Model CO line contrast against 4.5 μm continuum depth, with the observed cross-correlation significances. (d) The 2.1 μm–4.5 μm plane: the model trajectory couples the bands ($r = 0.96$) more tightly than observed ($r = 0.13$), because of Visits 2 and 5. (e) The same physics with a 17-day lag reproduces the 47-day CHEOPS period. (f) Robustness to stochastic outgassing bursts.

Table 3: Robustness of the band-contrast result to band definitions.

Window (μm)	CO_2 band (μm)	Faint Δ_C (ppm)	V4 Δ_C (ppm)	V1−V4 (ppm)
3.90–4.195	4.20–4.46	$+23.8 \pm 7.4$ (3.2σ)	-12.2 ± 8.2	$+38.3 \pm 12.3$ (3.1σ)
3.90–4.195	4.20–4.40	$+22.7 \pm 7.3$ (3.1σ)	-10.8 ± 8.8	$+40.2 \pm 13.0$ (3.1σ)
3.90–4.195	4.22–4.46	$+25.5 \pm 7.3$ (3.5σ)	-12.3 ± 8.4	$+39.5 \pm 12.4$ (3.2σ)
3.90–4.195	4.18–4.50	$+22.3 \pm 6.7$ (3.3σ)	-10.7 ± 7.8	$+37.3 \pm 11.7$ (3.2σ)
3.90–4.15	4.20–4.46	$+27.4 \pm 7.8$ (3.5σ)	-12.0 ± 8.6	$+41.9 \pm 13.2$ (3.2σ)
3.90–4.15	4.20–4.40	$+26.0 \pm 7.6$ (3.4σ)	-10.6 ± 9.2	$+43.9 \pm 13.7$ (3.2σ)
3.90–4.15	4.22–4.46	$+29.0 \pm 7.7$ (3.8σ)	-12.0 ± 8.8	$+43.1 \pm 13.2$ (3.3σ)
3.90–4.15	4.18–4.50	$+25.7 \pm 7.0$ (3.6σ)	-10.6 ± 8.3	$+41.0 \pm 12.5$ (3.3σ)
3.95–4.19	4.20–4.46	$+22.2 \pm 7.3$ (3.0σ)	-12.8 ± 8.4	$+36.9 \pm 12.4$ (3.0σ)
3.95–4.19	4.20–4.40	$+21.1 \pm 7.2$ (2.9σ)	-11.4 ± 8.9	$+38.7 \pm 13.0$ (3.0σ)
3.95–4.19	4.22–4.46	$+23.8 \pm 7.3$ (3.3σ)	-12.9 ± 8.6	$+38.0 \pm 12.4$ (3.1σ)
3.95–4.19	4.18–4.50	$+20.7 \pm 6.6$ (3.1σ)	-11.1 ± 8.0	$+35.8 \pm 11.7$ (3.1σ)